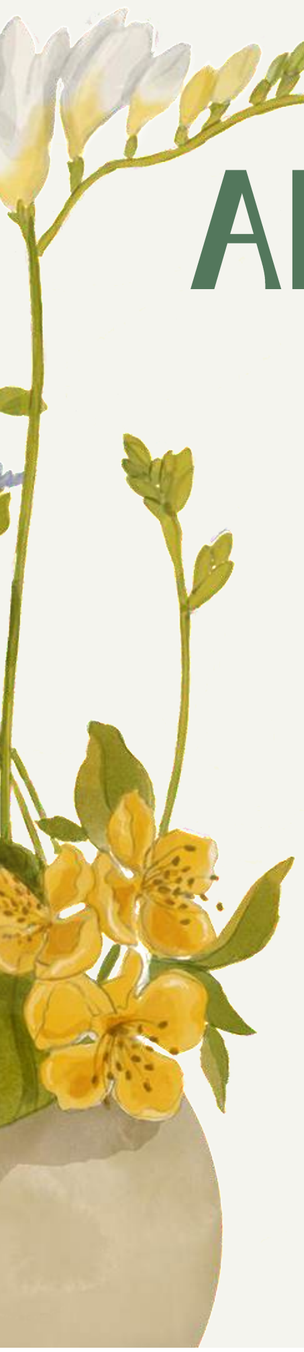
AI POWERED SMART ORCHID CARE SYSTEM USING MULTI-MODAL MACHINE LEARNING







AI Powered Smart Orchid Care System Using Multi-Modal Machine Learning





The Problem

Currently, no dedicated system exists for managing Vanda orchid farming. This leads to disease losses, poor plant management, and missed market opportunities.

Users and Stakeholders

- Commercial orchid growers
- breeders
- Exporters
- local flower markets

Impact

- Lost income from crop losses and wasted effort
- Solving can increase orchid grower's income



Existing Solutions and Their Limitations

- **OrchiZen:** IoT + ML but no severity assessment, no automated fertilization
- **Detection of Phalaenopsis Fusarium:** no severity grading or treatment advice
- **Watering systems:** no prediction, no nutrient forecasting
- **Multi-Modal Optical Sensing:** Focuses only on Phalaenopsis species, no continuous bloom prediction

Proposed Solution

- AI-powered integrated platform with four components
- Predictive care, data-driven decisions, confident harvest planning



Individual Contributions

Student ID	Name	Component
IT22065230	Wickramasinghe D.P	Hybrid pollination compatibility analysis
IT22064936	Rathnamalala D.M.J.P	Orchid Disease Detection, Severity Assessment and Treatment Recommendation
IT22062642	Vithanage P.V.P.M	Smart Watering and Automated Fertilization (Fertigation) for Vanda Orchids
IT22083500	Nimla Samadinee L.D	Orchid Growth Stage Classification & Bloom Prediction for Vanda Species











IT22065230 | Wickramasinghe D.P

Title: Hybrid pollination compatibility Analysis

Contribution: Predicts hybrid pollination compatibility between Vanda orchid parent plants using flower image analysis and machine learning. The component helps breeders identify suitable parent orchids and reduce the traditional trial-and-error process in orchid hybridization.

Novelty

Machine learning–based system to predict hybrid pollination compatibility in Vanda orchids by analysing flower morphological traits, reducing the traditional trial-and-error breeding process.

Problem Definition

Orchid breeders currently rely on manual knowledge and trial-and-error when selecting parent plants.Existing technologies focus on plant monitoring or flower classification, but do not predict hybrid pollination compatibility.

Research Objective & Sub-Objectives

Objective: Develop a machine learning-based system to analyse orchid flower traits and predict hybrid pollination success probability.

Sub-Objectives:

- Extract morphological features using image analysis
- Analyse compatibility between parent orchids using ML models
- Predict probability of successful hybrid pollination
- Provide recommended parent combinations



Latest Technologies Used to Develop the Proposed Solution



- Computer Vision :·Image processing techniques such as image segmentation and colour analysis to extract orchid flower traits.
- Image Feature Extraction :·Extraction of morphological features such as petal shape, symmetry, and texture patterns.
- Machine Learning Models :· Random Forest and Support Vector Machine (SVM) used for hybrid compatibility prediction.
- Feature Fusion:· Combines flower image features and parent plant characteristics for improved prediction accuracy.



How to Validate and Evaluate the Proposed Solution

- **Prediction Accuracy:** Compare predicted compatibility results with actual hybrid pollination outcomes recorded by orchid breeders.
- **Model Evaluation Metrics:** Evaluate model performance using Accuracy, Precision, Recall, and F1-score.
- **Cross-Validation:** Test the model on different Vanda orchid varieties to ensure reliable predictions.



Data Availability, Security and Ethical Clearance

- **Data Availability:** Orchid flower images and morphological traits collected from orchid growers and references such as **RHS Colour Chart classifications and orchid breeding records.**
- **Security:** Image datasets and compatibility prediction results stored in secure databases with controlled access.Ethical Clearance:
- **Ethical Clearance:** Permission obtained from orchid growers before collecting plant images or breeding data.



System Overview

- Camera module captures orchid flower images of potential parent plants
- Image processing analyses images to extract morphological traits (petal shape, colour, symmetry)
- Parent plant characteristics are combined with extracted image features
- Machine learning model predicts hybrid pollination compatibility probability
- Prediction results stored in the component database and shared with the main system

Dependencies and Common Components

- **Hardware:** Camera module for capturing orchid flower images
- **Software:** Python, OpenCV for image processing, machine learning libraries (Scikit-learn)
- **Common Components:** Shared system database, compatibility prediction module, central dashboard for displaying prediction results









IT22064936 | Rathnamalala D.M.J.P.

Title: Orchid Disease Detection, Severity Assessment & Treatment Recommendation for Vanda Species

Contribution: Automatically detects Phytophthora, Anthracnose, and Fusarium in Vanda orchids using plant images + environmental sensor data, then grades severity (early/moderate/severe) and provides treatment guidance + alerts to reduce plant loss and protect flower quality.

Novelty:

AI-based orchid disease monitoring for Vanda orchids, including severity assessment and severity-based treatment recommendation, beyond existing systems that mainly focus on disease detection only.

Problem Definition

Growers currently rely on manual inspection to identify diseases, which is slow and often inaccurate, especially in early stages. Most existing systems only detect disease presence and do not explain severity or recommend suitable treatments, leading to delayed action and plant loss.

Research Objective & Sub-Objectives

Objective: Build an AI-based system to automatically detect diseases in Vanda orchids, classify severity, and recommend suitable treatments.

Sub-Objectives:

- Detect Phytophthora, Anthracnose, and Fusarium from leaf, stem, and flower images
- Classify severity levels as early, moderate, and severe
- Generate treatment recommendations based on disease type and severity
- Display diagnosis results, severity levels, and alerts through a dashboard

Latest Technologies Used to Develop the Proposed Solution

- **Computer Vision:** Image-based disease recognition using deep learning models to identify orchid disease symptoms from plant images.
- **Severity Assessment (ML):** Multi-class classification to determine whether the infection level is early, moderate, or severe.
- **Multi-Modal Data Fusion:** Combines orchid image features and environmental sensor data
- **Treatment Recommendation:** Rule-guided and AI-assisted mapping from disease type + severity level to suggest suitable treatment actions.



How to Validate and Evaluate the Proposed Solution

- **Accuracy:** Compare predicted disease results with expert-labeled orchid disease samples.
- **Severity Classification Metrics:** Evaluate severity prediction for early, moderate, and severe classes
- **Treatment Relevance:** Validate recommended treatments with orchid growers or experts for correctness and safety.

Data Availability, Security and Ethical Clearance

- **Data Availability:** Images and sensor readings will be collected from greenhouse, covering healthy and diseased Vanda orchids
- **Security:** Images, sensor logs, and diagnosis results will be stored securely with controlled access.
- **Ethical Clearance:** Permission will be obtained from farm or nursery owners before data collection, and no personal human data will be involved.

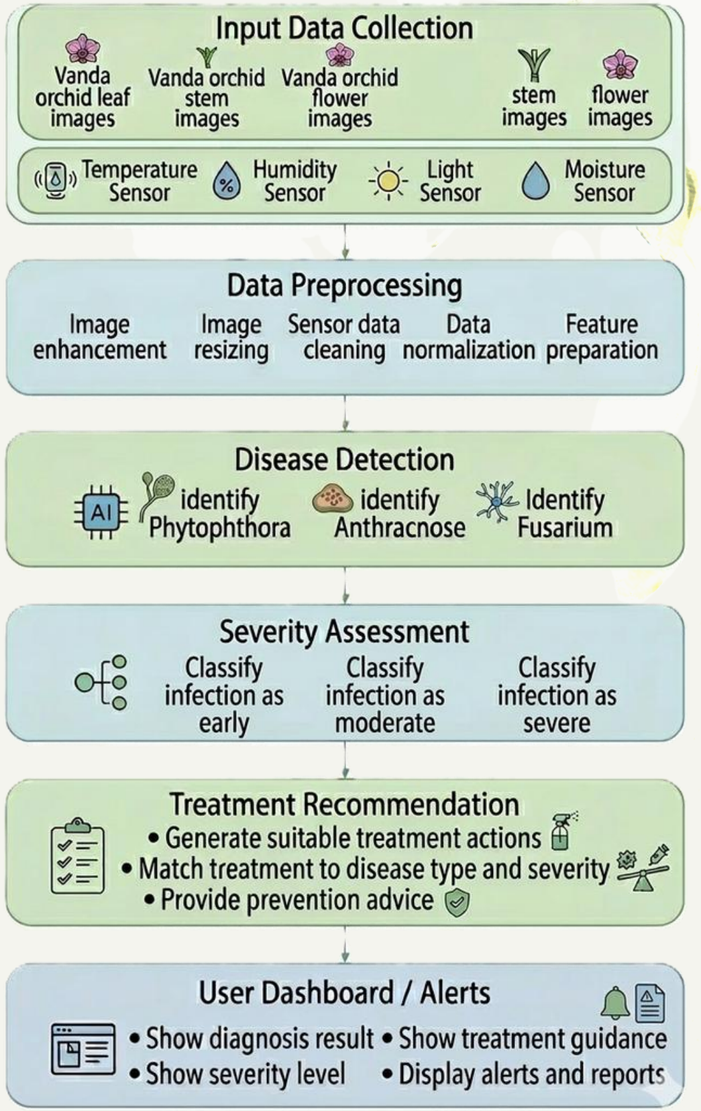


System Overview

- Capture orchid images together with sensor readings
- Preprocess images and clean sensor data for analysis
- AI model detects the disease type
- Severity assessment model classifies infection
- Generate suitable treatment recommendations
- Final results are stored and displayed through the central dashboard with history and reports

Dependencies and Common Components

- **Hardware:** Cameras, environment sensors
- **Software:** Python, image processing libraries
- **Common Components:** Data storage module , notification system for alerts, central dashboard for displaying predictions to farmers







IT22062642 | Vithanage P. V. P. M

Title: Smart Watering and Automated Fertilization System for Vanda Orchids

Contribution: Predicts watering requirements using environmental sensor data and machine learning. The system automatically controls irrigation and fertilization through water and fertilizer pumps, enabling efficient and precise care for Vanda orchids.

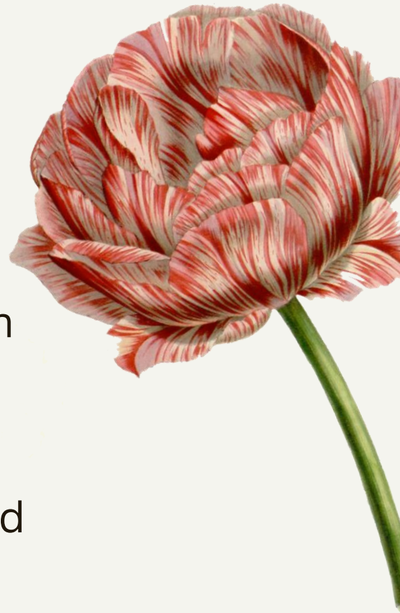
Smart Watering and Automated Fertilization System for Vanda Orchids

Novelty:

AI-based smart irrigation system designed specifically for Vanda orchids, addressing limitations of existing IoT irrigation systems that rely mainly on threshold-based watering and manual fertilization.

Problem Definition

- Vanda orchids require precise watering and nutrient control
- Farmers currently use manual schedules or visual observation
- Environmental changes (temperature, humidity, light) are not considered
- Existing research use threshold rules and many systems do not include automated fertilization



Research Objective & Sub-Objectives

Objective: Develop a smart irrigation and fertilization system using environmental sensor data and machine learning.

Sub-Objectives:

- Collect environmental data using IoT sensors
- Predict watering requirements using ML models
- Automate water and fertilizer pumps
- Provide monitoring via mobile dashboard



Latest Technologies Used to Develop the Proposed Solution

- **IoT Sensors:** Temperature, humidity, light, and moisture sensors to monitor environmental conditions.
- **Microcontroller:** ESP32 / Arduino used for sensor data collection and device control.
- **Machine Learning Models:** Random Forest, Decision Tree, and Logistic Regression for irrigation prediction.
- **Multi-Modal Data Fusion:** Combines environmental sensor data and historical irrigation data.
- **Time-Series Environmental Analysis:** Analyzes environmental patterns to improve watering predictions.





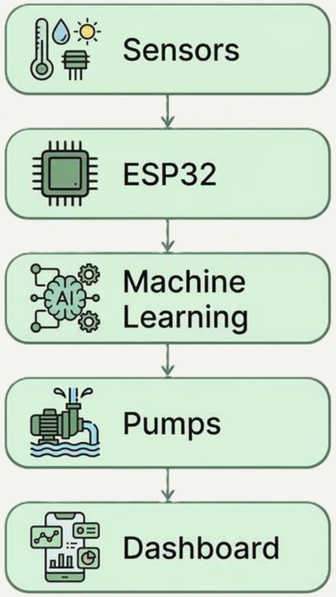
How to Validate and Evaluate the Proposed Solution

- **Accuracy:** Compare predicted watering decisions with expert grower recommendations.
- **Water Usage Efficiency:** Measure reduction in water usage compared with traditional irrigation methods.
- **Model Evaluation Metrics:** Evaluate model performance using Accuracy and Precision.



Data Availability, Security and Ethical Clearance

- **Data Availability:** Environmental sensor data collected from orchid farms.
- **Security:** Sensor data and predictions stored in secure databases with controlled access.
- **Ethical Clearance:** Permission obtained from farm owners before collecting environmental data.



System Overview

- Environmental sensors collect temperature, humidity, light, and moisture data
- Sensor data is processed and analyzed using machine learning models
- The system predicts watering requirements based on environmental conditions
- Automatically activates water pump and fertilizer pump
- Results are stored in the component database and shared with the main system

Dependencies and Common Components

- **Hardware:** Sensors, ESP32/Arduino microcontroller, water pump, fertilizer pump
- **Software:** Python, machine learning libraries (Scikit-learn / TensorFlow)
- **Common Components:** Shared system database, monitoring dashboard, alert notification system



IT22083500 | Nimla Samadinee L.D

Title: Orchid Growth Stage Classification & Bloom Prediction for Vanda Species
Contribution: Automatically identifies key growth stages and predicts flowering time to help farmers plan harvesting, delivery, and breeding activities.

Novelty:

AI-based growth stage monitoring for Vanda orchids, including post-bloom recovery tracking and bloom prediction, unlike existing systems designed only for Phalaenopsis species.

Problem Definition

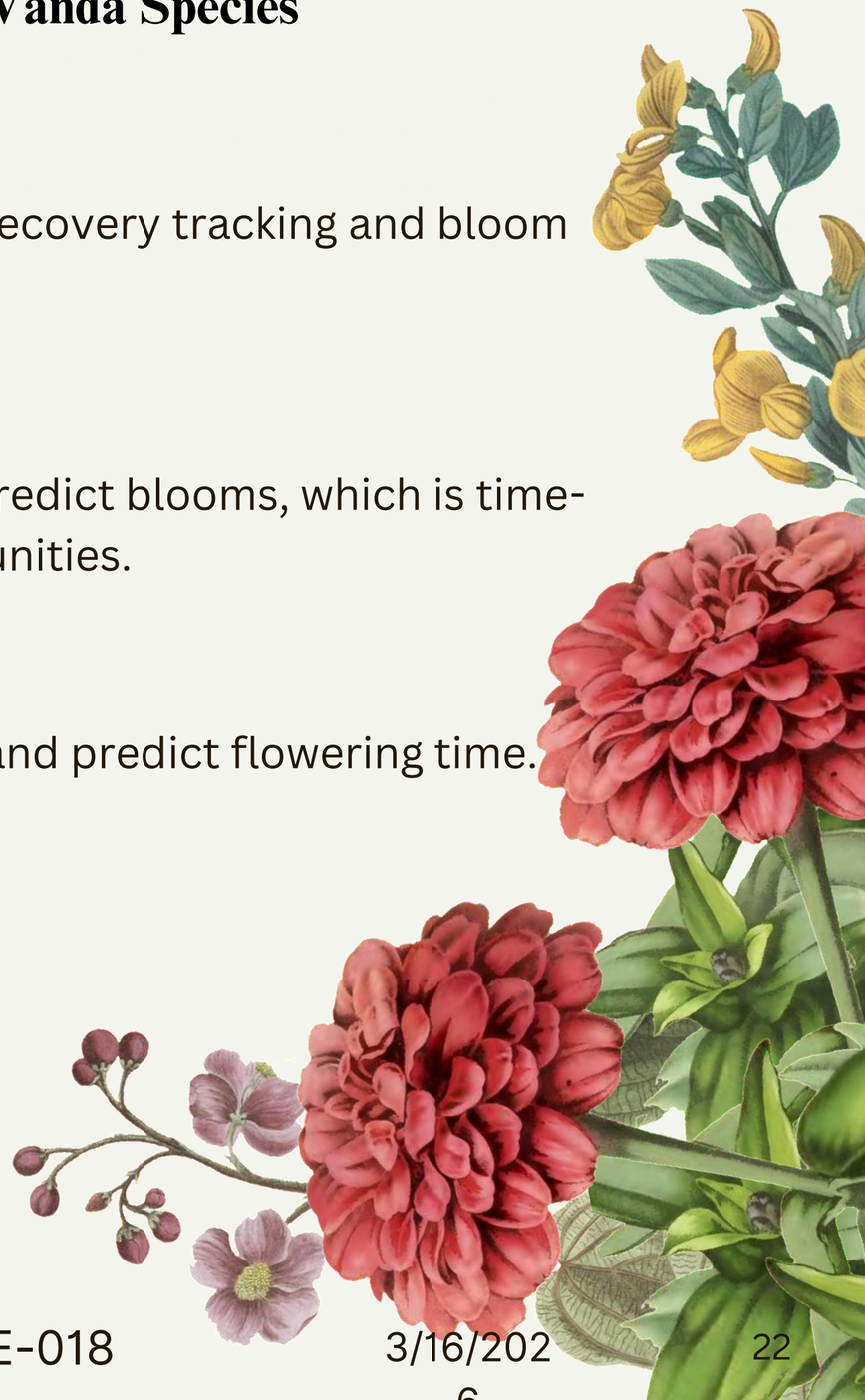
Farmers currently rely on manual inspection to identify growth stages and predict blooms, which is time-consuming , inaccurate, leading to poor planning and missed market opportunities.

Research Objective & Sub-Objectives

Objective: Build an AI model to automatically identify Vanda growth stages and predict flowering time.

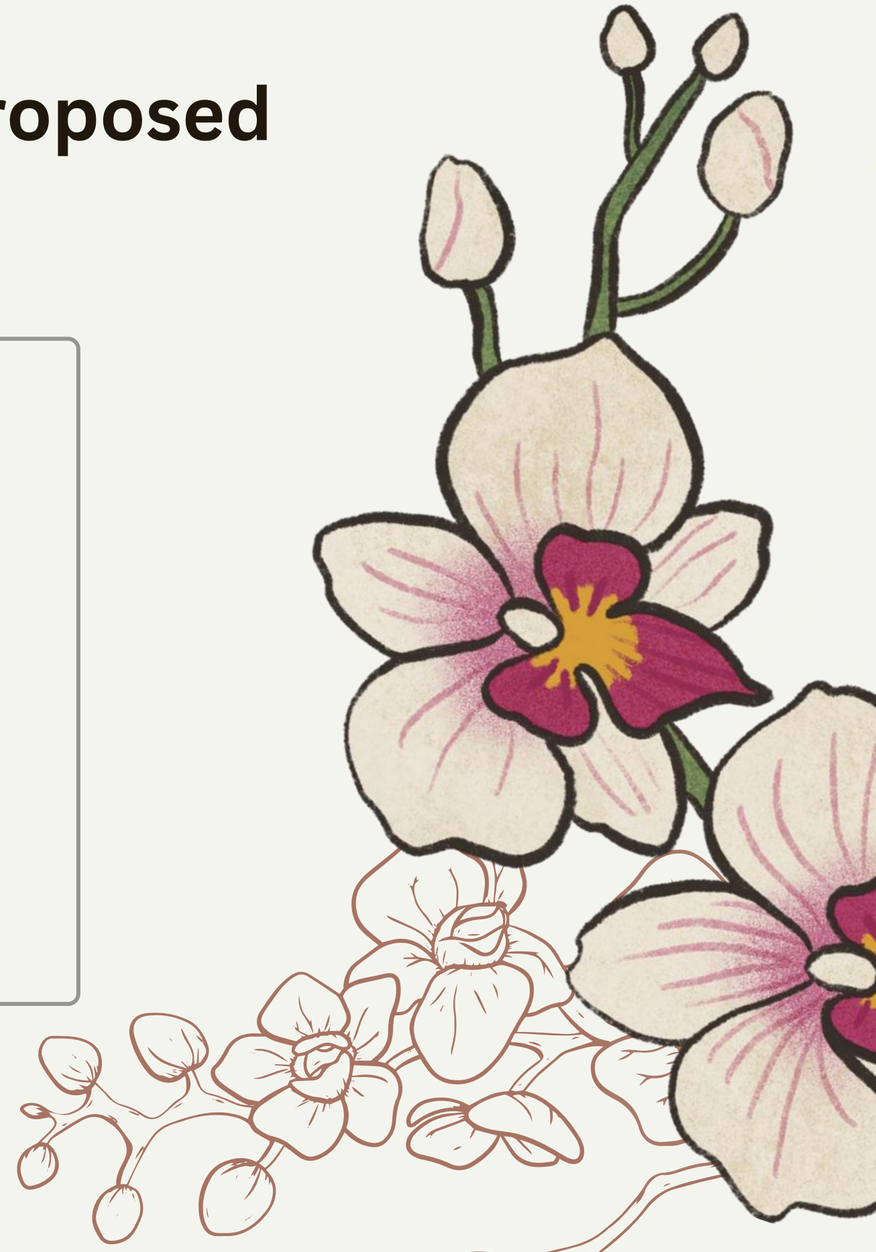
Sub-Objectives:

- Classify stages: seedling, vegetative, blooming, fading
- Predict days until flowering
- Estimate flower longevity
- Identify best time for seed collection



Latest Technologies Used to Develop the Proposed Solution

- **Deep Learning:**
  - **CNN:** For growth stage classification from images
  - **LSTM or Transformer Models:** For processing environmental data (temperature, humidity) over time
- **Multi-Modal Fusion Techniques:** To combine image and sensor features for final prediction
- **Time-Series Analysis:** Tracking growth patterns and environmental factors over time to make predictions



How to Validate and Evaluate the Proposed Solution

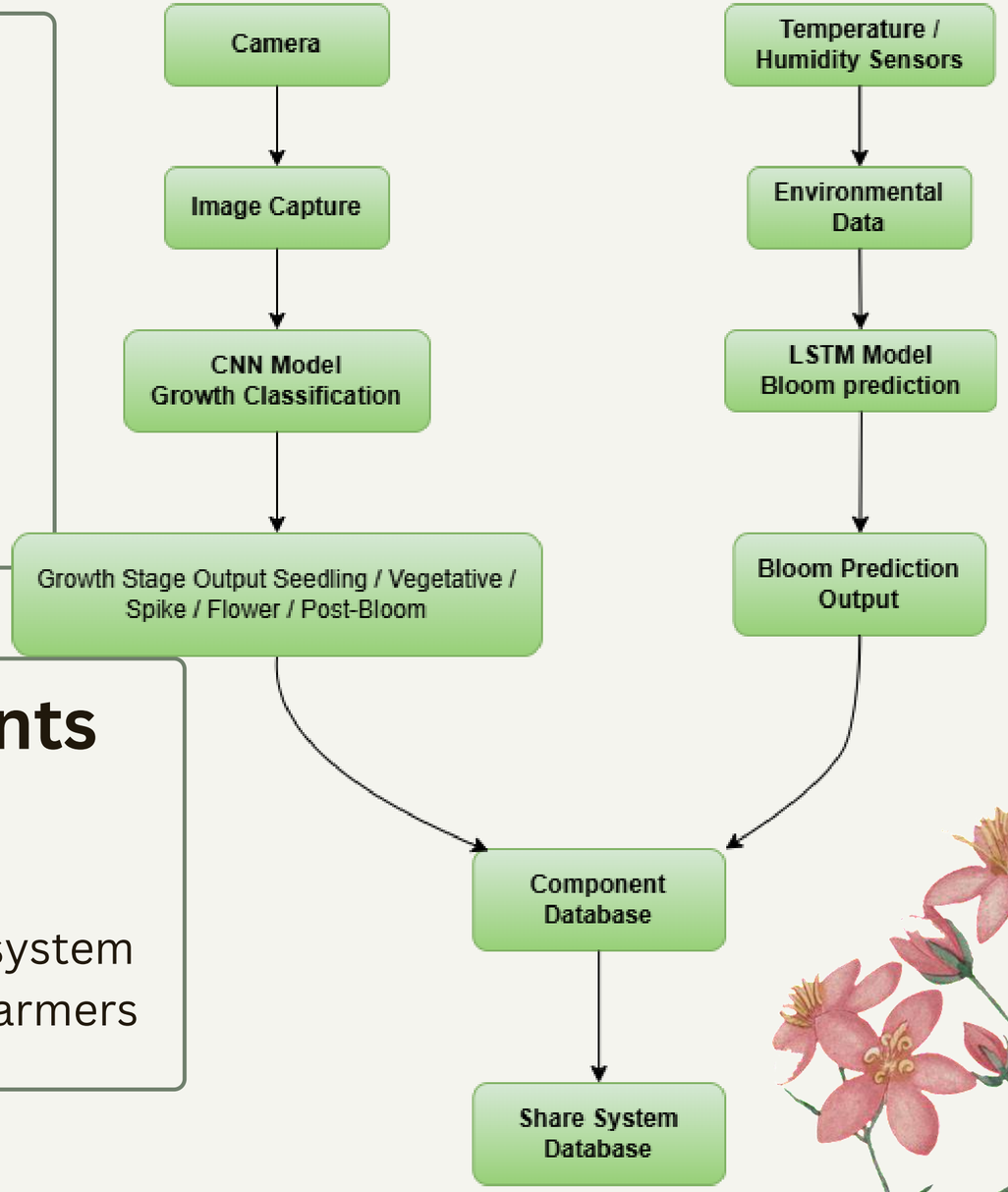
- **Accuracy:** Compare model's growth stage predictions against labels provided by orchid experts
- **Bloom Prediction Error:** Calculate the difference between predicted days to bloom and actual days to bloom
- **Cross-Validation:** Test the model on different Vanda hybrids under different growing conditions

Data Availability, Security and Ethical Clearance

- **Data Availability:** Will collect images and sensor data from local orchid farms over multiple growing cycles
- **Security:** Farm data will be stored locally or on secure servers with access controls; no sensitive personal data involved
- **Ethical Clearance:** Will obtain permission from farm owners before collecting data

System Overview

- Take daily photos and sensor readings (temp, humidity)
- CNN looks at photos to see plant features
- LSTM looks at sensor data to track environment changes
- Combine both to decide growth stage and predict bloom date
- Send result to main system to adjust care automatically



Dependencies and Common Components

- **Hardware:** Cameras, environment sensors
- **Software:** Python, image processing libraries
- **Common Components:** Data storage module , notification system for alerts, central dashboard for displaying predictions to farmers

AI Powered Smart Orchid Care System Using Multi-Modal Machine Learning

Ability of Commercialization

delivers one orchid-specific AI solution for disease detection, care forecasting, growth prediction, and pollination guidance to reduce plant loss and improve flower quality.

Investment Cost

- **Material:** Cameras, sensors, edge devices
- **Installation Labour:** Setup, training, maintenance
- **Total:** Estimated LKR 50,000 – 70,000 per greenhouse

How to Recover Investment

- Sell hardware + software as one-time package
- Monthly subscription
- Charge for premium features

Benefits of Our System

Saves time, reduces manual errors, helps farmers plan better, improves flower quality

Target Market:

Commercial orchid farms, nurseries, exporters.
Farm owners and greenhouse managers.

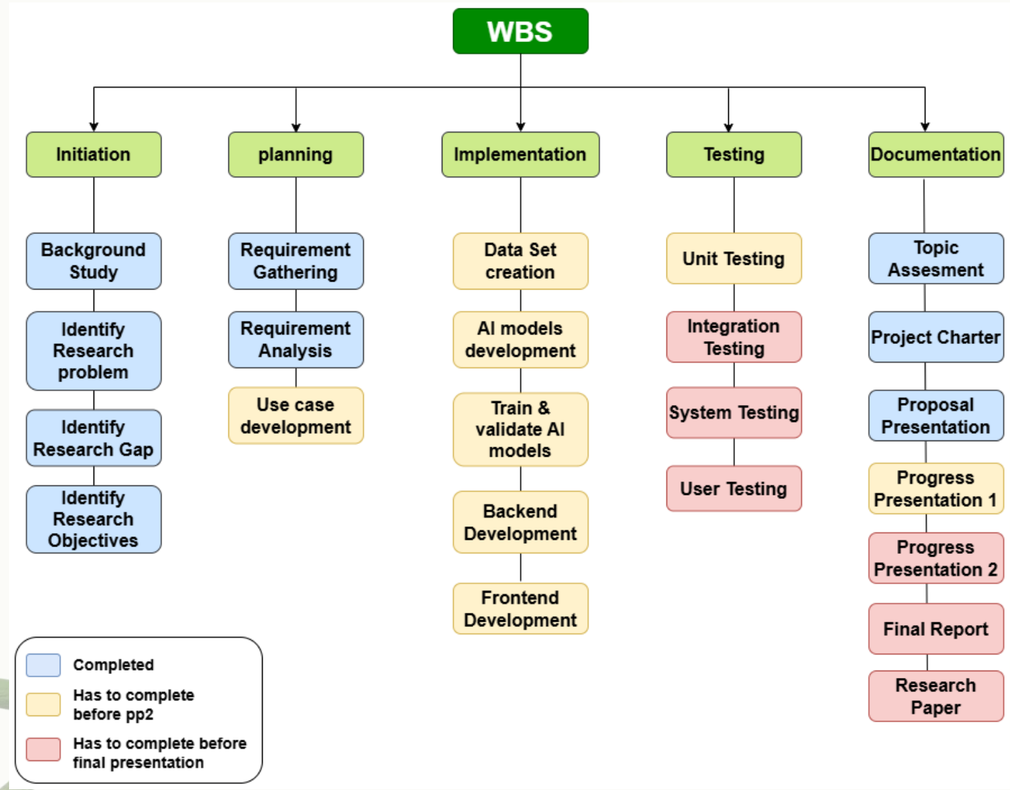
Intellectual Property Rights

Copyright for software, patent for system design, trademark for branding.

Gantt Chart

	Semester 1					Semester 2					
Activity	1	2	3	4	5	6	7	8	9	10	11	12
Preparation												
Identify the Problem												
Requirement gathering												
Literature review												
Analyze requirements												
Online project registration												
Topic assessment from submission												
Identify main objectives												
Project proposal submission												
Data Collection												
Model Training												
Software Development												
Progress presentation 1												
Integration												
Testing												
Progress presentation 2												
Research paper submission												
Progress presentation 2												
Finalizing												

Work breakdown structure




Thanks